Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	comments
FR-001	Functional	The system shall match the driver vehicle connector type (CCS, CHAdeMO, Type 2) with available station chargers and reserve a 30-minute charging window.	High	**Pass:** Incompatible connector plugs are hidden from the driver. **Fail:** Two vehicles are booked for the same socket concurrently.	Ensures target hardware compatibility and prevents booking collisions on physical charging sockets.	Clear functional scope; need to ensure backend handles concurrency locks to avoid race conditions.
FR-002	Functional	The system shall display real-time charger availability and calculate dynamic power rates based on current grid demand and time slot.	High	**Pass:** Power rate updates dynamically on screen during rate changes. **Fail:** Incorrect pricing or unavailable chargers displayed as operational.	Provides price transparency and allows drivers to locate active, compatible stations.	State update refresh rate (e.g., polling vs WebSockets) should be specified for real-time accuracy.
FR-003	Functional	The system shall allow EV drivers to navigate to the selected charging station using built-in location services.	Medium	**Pass:** System routes driver from current location to station GPS coordinates. **Fail:** GPS routing fails or provides invalid coordinates.	Assists drivers in locating reserved sockets efficiently.	Specify third-party navigation API integration (e.g., Google Maps API or Mapbox).
FR-004	Functional	The system shall process digital payments to confirm and lock a reservation time slot	High	**Pass:** Reservation status changes to "Confirmed" upon payment gateway authorization. **Fail:** Slot booked without payment authorization.	Secures revenue for the station operator and prevents unpaid slot hoarding.	Clarify payment gateways to support (e.g., Credit Card, UPI, Mobile Wallet) and

						timeout window.
FR-005	Functional	The system shall allow station operators to manually override station socket status (e.g., set to maintenance or offline).	Medium	**Pass:** Disabled socket immediately removes availability from EV driver search views. **Fail:** Reserved socket permits driver booking while marked offline.	Enables operators to maintain infrastructure without risking double-bookings or driver inconvenience.	Add behavior for existing bookings when a socket is suddenly marked offline (e.g., auto-refund/re-route).
NFR-001	Non-Functional (Performance & Security)	The system shall automatically release locked sockets and apply a penalty if the driver fails to plug in within 10 minutes of slot start.	High	**Pass:** Socket releases and penalty charge posts after 10 minutes idle. **Fail:** Socket remains locked past 10 minutes without plug-in detection.	Maximize station utilization and deter slot holding by no-show drivers.	Requires reliable physical sensor integration/OCPP protocol integration at the station to detect plug-in event.
NFR-002	Non-Functional (Performance & Security)	The system shall encrypt all transaction data using AES-256 and maintain system availability of 99.9% during operational hours.	High	**Pass:** Penetration testing confirms data encryption; uptime logs verify ≥99.9% availability. **Fail:** Plaintext data exposure or service downtime exceeds 0.1%.	Protects sensitive user financial data and ensures continuous availability for drivers.	Meets standard security compliance; uptime calculation window (monthly/yearly) should be formally defined.